GATE ME 90 Days Challenge

Day 48/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Weekly Revision	Week 8 Revision	56 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on a mixed revision of the week: fins and lumped transient conduction, forced and free convection, permutations, probability, mensuration and data interpretation, radiation exchange and heat exchangers, and gas and vapour power cycles. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE.
Today's test: open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A fin is found to have an effectiveness of 0.8. This means that

(A) the fin is 80 % efficient

(B) the fin actually reduces the heat transfer from the surface and acts as insulation

(C) the fin removes 80 % more heat than the bare surface

(D) the fin is too short

Q2 NAT

An aluminium sphere of 10 mm diameter (ρ = 2700 kg/m³, c = 900 J/kg·K, k = 200 W/m·K) at 250 °C is cooled in air at 30 °C with h = 60 W/m²·K. The time taken for the sphere to cool to 100 °C (in s, round off to one decimal place) is _____

Q3 NAT

A very long copper rod (k = 380 W/m·K) of 3 mm diameter projects from a wall 60 °C above the surrounding air; h = 25 W/m²·K. The heat transfer from the rod (in W, round off to two decimal places) is _____

Q4 MCQ

The Stanton number used in convection is

(A) Nu × Re × Pr

(B) Re/(Nu Pr)

(C) hL/k

(D) Nu/(Re Pr) = $h/(\rho V c_p)$

Q5 NAT

For fully developed laminar flow in a circular tube with a uniform wall temperature, Nu = 3.66. Water (k = 0.6 W/m·K) flows in a tube of 10 mm diameter. The heat transfer coefficient (in W/m²·K, round off to one decimal place) is _____

Q6 MCQ

For turbulent flow over a flat plate (local $Nu_x = 0.0296\ Re_x^{0.8} Pr^{1/3}$), the local heat transfer coefficient h_x varies with distance x from the leading edge as

(A) $x^{-0.5}$
(B) $x^{0.2}$
(C) $x^{-0.2}$
(D) independent of x

Q7 MCQ

Two cards are drawn one after the other without replacement from a well-shuffled pack of 52 cards. The probability that both are aces is

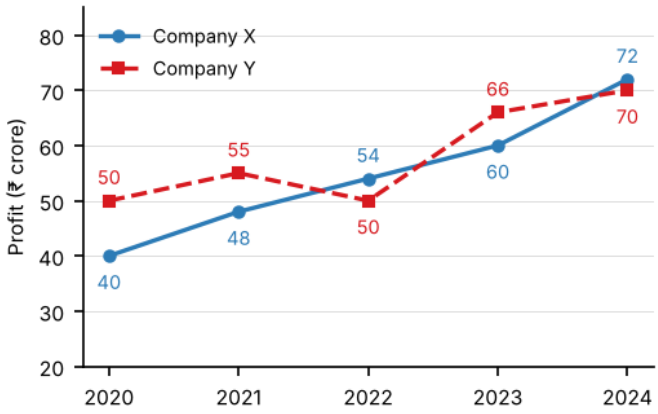
(A) 1/169
(B) 1/221
(C) 1/26
(D) 2/221

Q8 NAT

Ten points lie in a plane, of which exactly 4 are collinear and no other three are collinear. The number of triangles that can be formed with these points as vertices is _____

Q9 MCQ

The line graph shows the annual profit (₹ crore) of companies X and Y from 2020 to 2024. By about what percentage did the combined profit of the two companies increase from 2020 to 2024?



(A) 36.6 %
(B) 44.0 %
(C) 52.0 %
(D) 57.8 %

Q10 MCQ

Inside a closed cubical enclosure, the shape factor from one face to each adjacent face is 0.2. The shape factor from a face to the opposite face is

(A) 0.2
(B) 0.25
(C) 0.167
(D) 0.8

Q11 NAT

In a counter-flow heat exchanger of 5 m² area (U = 400 W/m²·K), the hot fluid cools from 160 °C to 100 °C and the cold fluid is heated from 40 °C to 90 °C. The heat transfer rate (in kW, round off to one decimal place) is _____

Q12 MCQ

In an evaporator, the refrigerant boils at constant temperature while air flows over the tubes. If the exchanger has NTU = 2, its effectiveness is about

(A) 0.500
(B) 0.667
(C) 0.865
(D) 0.981

Q13 MCQ

In a Rankine cycle the condenser pressure is lowered, keeping the boiler conditions unchanged. Then

(A) the efficiency increases but the steam leaving the turbine becomes wetter
(B) the efficiency decreases
(C) the turbine exhaust becomes drier
(D) the pump work decreases

Q14 NAT

A gas turbine plant with a regenerator works on air with constant c_p. The actual temperatures are: compressor inlet 300 K, compressor exit 500 K, turbine inlet 1100 K and turbine exit 700 K. The regenerator effectiveness is 0.75. The thermal efficiency of the plant (in %, round off to two decimal places) is _____

Q15 NAT

An ideal Brayton cycle on air ($\gamma = 1.4$) works between 300 K and 1350 K. The pressure ratio that gives the maximum net work per kg (round off to two decimal places) is _____

Answer Key

Q	Type	Answer
Q1	MCQ	(B) the fin actually reduces the heat transfer from the surface and acts as insulation
Q2	NAT	77.3 (accept 76.8 to 77.8)
Q3	NAT	1.51 (accept 1.49 to 1.53)
Q4	MCQ	(D) $Nu/(Re\ Pr) = h/(\rho V c_p)$
Q5	NAT	219.6 (accept 218.5 to 220.7)
Q6	MCQ	(C) $x^{-0.2}$
Q7	MCQ	(B) 1/221
Q8	NAT	116
Q9	MCQ	(D) 57.8 %
Q10	MCQ	(A) 0.2
Q11	NAT	129.7 (accept 129.0 to 130.4)
Q12	MCQ	(C) 0.865
Q13	MCQ	(A) the efficiency increases but the steam leaving the turbine becomes wetter
Q14	NAT	44.44 (accept 44.10 to 44.80)
Q15	NAT	13.90 (accept 13.80 to 14.00)

Detailed Solutions

Q1 **Answer: (B) the fin actually reduces the heat transfer from the surface and acts as insulation**

Concept: Effectiveness ε_f = (heat transfer with fin)/(heat transfer from the base area without the fin).

$\varepsilon_f < 1$ means the fin conducts less heat than the bare area it covers: it **acts as insulation (B)**.

This happens when h is high or k is low ($hA_c/(kP)$ not small). A fin is normally justified only when $\varepsilon_f \geq 2$.

Exam tip: Do not confuse effectiveness with efficiency (actual heat/heat if the whole fin were at base temperature).

Q2 **Answer: 77.3 (accept 76.8 to 77.8)**

Concept: Check Bi; if Bi < 0.1 use lumped capacitance: $(T - T_\infty)/(T_i - T_\infty) = e^{-t/\tau}$, $\tau = \rho c L_c/h$.

$L_c = D/6 = 1.667$ mm; $Bi = 60 \times 0.001667/200 = 0.0005 \ll 0.1$

$\tau = 2700 \times 900 \times 0.001667/60 = 67.5$ s

$t = 67.5 \times \ln(220/70) = 67.5 \times 1.1451 =$ **77.3 s**

Exam tip: For a sphere $L_c = R/3 = D/6$, not R.

Q3 **Answer: 1.51 (accept 1.49 to 1.53)**

Concept: Infinitely long fin: $Q = \sqrt{(hPkA_c)}\,\theta_b$.

$P = \pi \times 0.003 = 9.425 \times 10^{-3}$ m; $A_c = \pi(0.003)^2/4 = 7.069 \times 10^{-6}$ m²

$hPkA_c = 25 \times 9.425 \times 10^{-3} \times 380 \times 7.069 \times 10^{-6} = 6.329 \times 10^{-4}$; $\sqrt{} = 0.02516$ W/K

$Q = 0.02516 \times 60 =$ **1.51 W**

Exam tip: $m = \sqrt{(4h/(kD))} = 9.37\ m^{-1}$; the long-fin formula holds when mL > about 2.65 (L > 0.28 m here).

Q4 **Answer: (D) $Nu/(Re\,Pr) = h/(\rho Vc_p)$**

Concept: $St = h/(\rho Vc_p)$: heat actually transferred to the wall relative to the heat-capacity flux of the stream.

Since $Nu = hL/k$, $Re = \rho VL/\mu$ and $Pr = \mu c_p/k$, $Nu/(Re\,Pr) = h/(\rho Vc_p)$: **(D)**.

It appears in the Reynolds–Colburn analogy $St\,Pr^{2/3} = C_f/2$.

Exam tip: Peclet number = Re × Pr, so St = Nu/Pe.

Q5 **Answer: 219.6 (accept 218.5 to 220.7)**

Concept: In fully developed laminar flow Nu is a constant: 3.66 (uniform wall temperature) or 4.36 (uniform heat flux).

$h = Nu\,k/D = 3.66 \times 0.6/0.01 =$ **219.6 W/m²·K**

It does not depend on the velocity at all, only on D.

Exam tip: Halving D doubles h in laminar flow: one reason for compact heat exchangers with narrow passages.

Q6 **Answer: (C) $x^{-0.2}$**

Concept: $h_x = Nu_x\,k/x$.

$Nu_x \propto x^{0.8}$, so $h_x \propto x^{0.8}/x =$ **$x^{-0.2}$ (C)**.

In laminar flow, $Nu_x \propto x^{0.5}$ gives $h_x \propto x^{-0.5}$: h falls off much faster.

Exam tip: At transition h_x jumps up sharply because turbulence thins the viscous sublayer.

Q7 **Answer: (B) 1/221**

Concept: Without replacement, multiply the conditional probabilities.

$P = (4/52) \times (3/51) = 12/2652 =$ **1/221 (B)**

Same as C(4, 2)/C(52, 2) = 6/1326.

Exam tip: With replacement it would be $(1/13)^2 = 1/169$, option (A).

Q8 **Answer: 116**

Concept: Every choice of 3 non-collinear points gives a triangle.

All selections: C(10, 3) = 120

Collinear selections from the 4 points: C(4, 3) = 4 (these give no triangle)

Triangles = 120 − 4 = **116**

Exam tip: For straight lines through pairs of these points: C(10, 2) − C(4, 2) + 1 = 40.

Q9 **Answer: (D) 57.8 %**

Concept: Read the values, add, then percentage change = (new − old)/old × 100.

2020: X = 40, Y = 50, total 90. 2024: X = 72, Y = 70, total 142.

Increase = 52/90 × 100 = **57.8 % (D)**

Exam tip: 36.6 % = 52/142 uses the wrong base; always divide by the earlier value.

Q10 **Answer: (A) 0.2**

Concept: Summation rule: the shape factors from a surface to all surfaces of an enclosure add up to 1.

A flat face does not see itself: $F_{11} = 0$. It has 4 adjacent faces and 1 opposite face.

$4 \times 0.2 + F_{opp} = 1 \rightarrow F_{opp} =$ **0.2 (A)**

Exam tip: So each face sees every other face equally: 1/5 each, as symmetry also suggests.

Q11 **Answer: 129.7 (accept 129.0 to 130.4)**

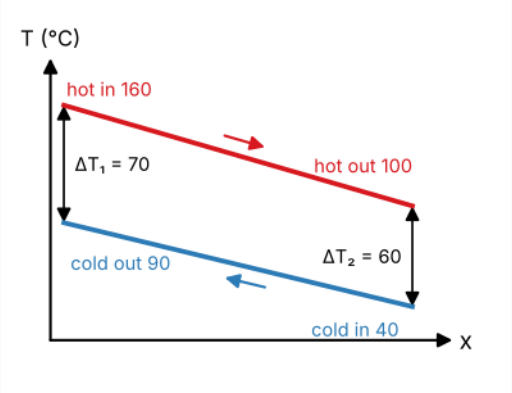
Concept: Q = UA × LMTD with the terminal differences for counter-flow.

$\Delta T_1 = 160 - 90 = 70$ °C (hot in, cold out); $\Delta T_2 = 100 - 40 = 60$ °C

LMTD = (70 − 60)/ln(70/60) = 10/0.1542 = 64.87 °C

Q = 400 × 5 × 64.87 = **129.7 kW**

Exam tip: When $\Delta T_1/\Delta T_2 < 1.5$, the arithmetic mean (65 °C) is within 0.3 % of the LMTD.



Q12 **Answer: (C) 0.865**

Concept: A boiling or condensing fluid has C → ∞, so $C_r = 0$.

For $C_r = 0$ every arrangement gives $\varepsilon = 1 - e^{-NTU}$.

$\varepsilon = 1 - e^{-2} = 1 - 0.135 =$ **0.865 (C)**

Exam tip: 0.667 = NTU/(1 + NTU) is the counter-flow value for $C_r = 1$; 0.5 is the parallel-flow limit for $C_r = 1$.

Q13 **Answer: (A) the efficiency increases but the steam leaving the turbine becomes wetter**

Concept: Lower condenser pressure lowers the temperature at which heat is rejected.

The net work rises and the mean temperature of heat rejection falls, so η **increases**.

But the expansion line extends further into the wet region: the exhaust is **wetter (A)**. Pump work increases slightly.

Exam tip: The lowest condenser pressure is set by the cooling-water temperature (about 5–10 kPa).

Q14 **Answer: 44.44 (accept 44.10 to 44.80)**

Concept: The regenerator heats the compressed air from T_2 to $T_x = T_2 + \varepsilon(T_4 - T_2)$, so the fuel supplies only $c_p(T_3 - T_x)$.

$T_x = 500 + 0.75 \times (700 - 500) = 650$ K

$w_{net}/c_p = (1100 - 700) - (500 - 300) = 200$ K; $q_{in}/c_p = 1100 - 650 = 450$ K

$\eta = 200/450 =$ **44.44 %** (without regenerator: 200/600 = 33.3 %)

Exam tip: The regenerator changes only the heat input, never the net work.

Q15 **Answer: 13.90 (accept 13.80 to 14.00)**

Concept: Maximum net work when $T_2 = T_4 = \sqrt{(T_1 T_3)}$, i.e. $r_p = (T_3/T_1)^{\gamma/(2(\gamma-1))}$.

$T_3/T_1 = 1350/300 = 4.5$; $\gamma/(2(\gamma - 1)) = 1.75$

$r_p = 4.5^{1.75} =$ **13.90**

Exam tip: At this pressure ratio, $T_2 = T_4 = 636.4$ K, so a regenerator would be of no use.

What next?

Day 48 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

Website test series

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Next, Day 49/90: Applied Thermodynamics, IC Engines. Comment **GATE90** on our reel to get all links. Want concise revision notes? The GATE ME handwritten short notes of an AIR 16 topper are available in the Dig Career Thrust app.